CANDIDATE NAME	CENTER NUMBER	CANDIDATE NUMBER
	0 6 0 1 5 0	

For Performance Measurement

ZIMBABWE SCHOOL EXAMINATIONS COUNCIL
General Certificate of Education Ordinary Level

COMBINED SCIENCE **4003/2**
PAPER 2 Theory
NOVEMBER 2021 SESSION **2 hours**

Learners answer on question paper
Additional material: Electronic calculator
Pencil (type HB recommended)

The Periodic Table is provided on page 10

TIME 2 hours

INSTRUCTIONS TO CANDIDATES
Write your name, centre number and candidate number in the spaces at the top of this page

SECTION A
Answer all questions in the spaces provided

SECTION B
Answer ***three*** questions from this section.

SECTION C
Answer ***three*** questions from this section.

SECTION D
Answer ***three*** questions from this section

INFORMATION FOR CANDIDATES
The number of marks is shown in brackets [ ]

EXAMINER'S USE	
A	
SECTION B	
SECTION C	
SECTION D	
TOTAL	

This document consists of 9 printed pages

 [Turn over

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Section A

Answer all questions in this section in the spaces provided.

1. **Fig. 1.1** shows a pyramid of biomass.

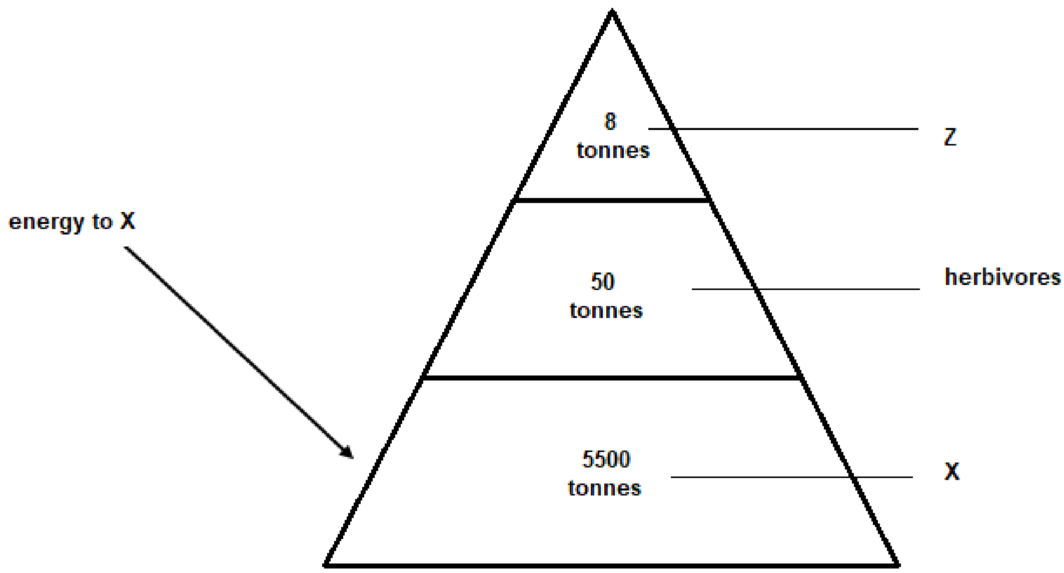

Fig. 1.1

a. Define the term ***biomass***

....[1]

b. i. Identify the trophic level represented by **X.**

....[1]

ii. Explain the shape of the pyramid.

....

....

....[2]

iii. Name the form of energy received by **X.**

....[1]

iv. State, giving a reason, the effect of decreasing the biomass of **X** on **Z**.

....

....[2]

[Total: 7]

2. a. Write a word equation for anaerobic respiration in mammals.

....

....[2]

b. Describe how plants are adapted to reduce water loss due to transpiration.

....

....

....[2]

c. Relate the structure of a blood capillary to its function.

....

....

....

....[2]

[Total: 6]

3. Fig. 3.1 shows the electrolysis of molten lead (II) bromide.

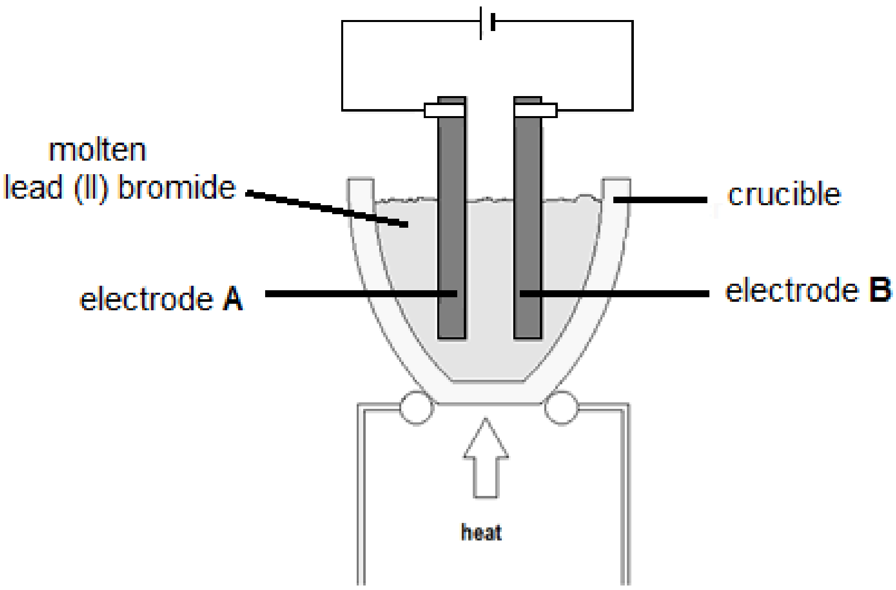

Fig. 3.1

a. i. Define the term electrolysis.

……

……[1]

ii. State the product formed at each of the electrodes **A** and **B**

A. ………………………………………………………………………………………………..

B. ……………………………………………………………………………………………… [2]

C.

b. i. Suggest the most suitable material which can be used as electrodes.

……[1]

ii. Give two general properties of an electrode.

1. ………………………………………………………………………………………………..

………………………………………………………………………………………………..

2. ………………………………………………………………………………………………..

………………………………………………………………………………………………..[2]

c. State any one reason for plating iron.

……[1]

[Total: 7]

4. a. Two reactions **A** and **B** are shown below.

*Reaction **A***

iron + *sulphur* $\longrightarrow$ *iron sulphide*

*Reaction **B***

sugar + *water* $\longrightarrow$ *sugar solution*

i. Name a process which can be used to obtain pure water from the sugar solution.

……[1]

ii. State the reaction, A or B, in which there is a physical change.

…… [1]

iii. State any two factors that affect solubility.

1. …………………………………………………………………………………………………..
2. …………………………………………………………………………………………………..[2]

b. Describe the particles in a gas.

………

………

………[2]

[Total: 6]

5. a. i. State the law of conservation of energy.

…………………………………………………………………………………………………….

…………………………………………………………………………………………………….[2]

ii. Write the energy conversion in a stretched catapult.

…………………………………………………………………………………………………….

…………………………………………………………………………………………………….[2]

b. Describe the Hwange Thermal Power generation.

………

………

………[3]

[Total: 7]

6. a. i. State the formula for calculating pressure in liquids.

…………………………………………………………………………………………………….[1]

ii. Calculate the pressure exerted by a 1.5 m column of water given that its density is 1200kgm^{-3}
[acceleration due to gravity = 10ms^{-2}]

[2]

b. Fig. 6.1 shows a model of a siphon being used to drain a liquid.

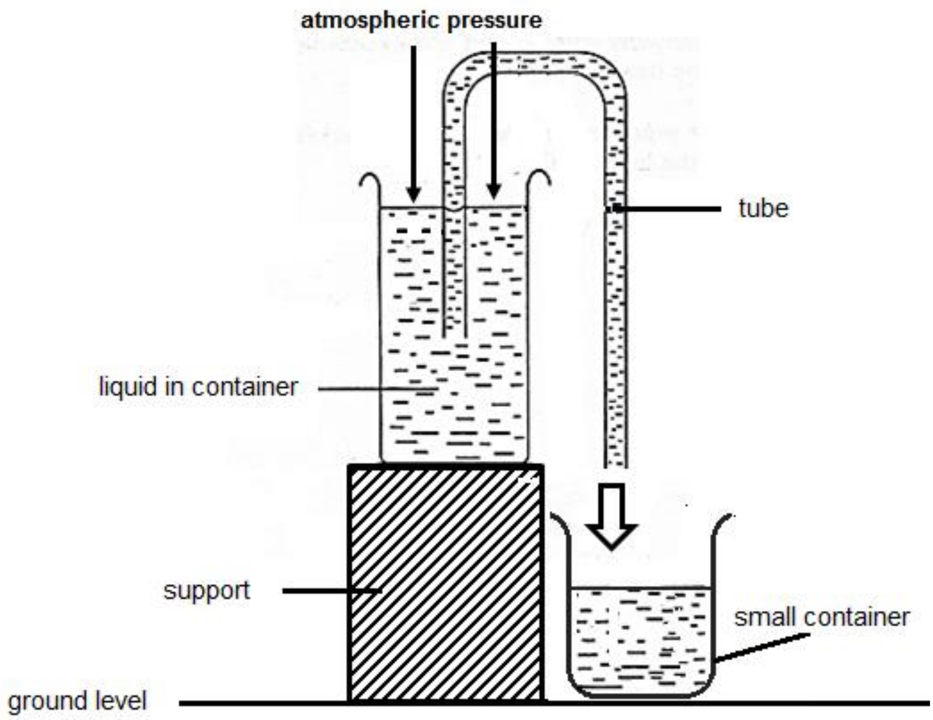

Fig. 6.1

i. Give two conditions that enable the siphon to work.

1. ……………………………………………………………………………………………..

2. ……………………………………………………………………………………………..[2]

ii. Describe how the siphon works.

…………………………………………………………………………………………………….

…………………………………………………………………………………………………….

…………………………………………………………………………………………………….

…………………………………………………………………………………………………….[2]

[Total: 7]

Section B: Biology

Answer any **two** questions on the separate answer sheets provided.

7. a. Describe any **three** ways that can be used to dispose of household litter. [3]

b. i. Outline the importance of maintaining clean toilets at school. [3]

ii. List any **three** materials or substances that can be used for cleaning the toilets. [3]

c. Describe any **one** role of the Environmental management Agency (EMA). [1]

[Total: 10]

8. a. state the function of the

1. testes,
2. Sperm duct,
3. Prostate gland. [3]

b. i. Describe the lifecycle of the bilharzia parasite. [4]

ii. State three methods of preventing bilharzia. [3]

[Total: 10]

9. a. i. Define the term balanced diet. [1]

ii. State any one function of fibre in the diet. [1]

iii. Name any two sources of protein for a person who does not eat meat. [2]

b. Plan a meal for a manual worker's lunch indicating the nutrients provided. [4]

c. State two ways by which Ebola is transmitted in a community. [2]

[Total: 10]

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Section C : Chemistry

Answer any **two** questions on the separate answer sheets provided.

10. The electronic configuration for three elements **X**, **Y** and **Z** are

X 2, 8, 6
Y 2, 1
Z 2, 6

a. i. State, using **X**, **Y** or **Z**, the two elements that are in the same group of the Periodic Table. [2]
ii. Give a reason for the answer in **a. i.** [1]
iii. Identify, from **X**, **Y** and **Z**, the element that has the highest proton number. [1]
iv. Name the type of bonding that can exist between **Y** and **Z**. [1]

b. i. **Z** is an isotope and it has eight neutrons.
Define the term isotope. [1]
ii. State the nucleon number of **Z**. [1]

c. Describe three differences in the physical properties of **Y** and **Z**. [3]
[Total: 10]

11. a. i. Define the term fuel. [1]
ii. State any two uses of fuels. [2]
iii. State any three alternative sources of energy other than fuels. [3]

b. i. Name the gas that causes global warming. [1]
ii. State any three effects of global warming. [3]
[Total: 10]

12. a. Define the term *neutralization.* [1]

b. State two formula that may be used to calculate the concentration of a solution. [2]

c. i. Name the process of making soap. [1]
ii. Describe how soap is produced from vegetable oil. [4]
iii. State the second product of r the process named. [1]
[Total: 10]

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Section D : Physics

Answer any **two** questions on the separate answer sheets provided.

13. a. i. A 2A electric heater was connected to a 110V supply for 1 hour.
Calculate the cost of running the electric heater for 1 hour if one unit of electricity costs 50 cents. [4]

ii. State one limitation of Ohms' law. [1]

iii. Name the two cables on a two pin plug. [2]

b. i. Outline how a lightning conductor should be installed for it to protect a building. [2]

ii. State any one myth about lightning. [1]

[Total: 10]

14. a. i. Describe the operation of a direct current (d.c) motor. [5]

ii. State any three factors that affect the speed of rotation of the coil. [3]

b. State any two uses of solar systems. [2]

[Total: 10]

15. a. A boy pushes a wheel barrow with a force of 25N against a frictional force of 7N.

i. Define the term *friction.* [1]

ii. Calculate the resultant force on the wheel barrow. [2]

b. State any two applications of friction. [2]

c. Fig. 15.1 shows a borehole which is operated by a lever. The load is 120N.

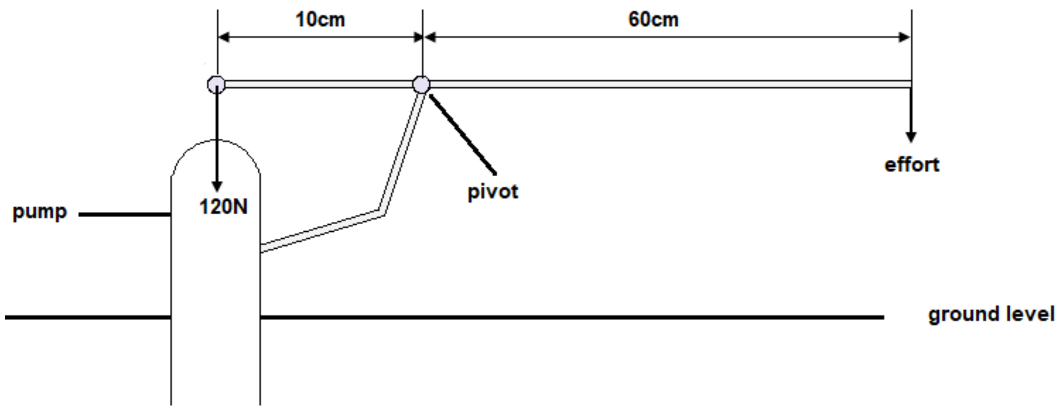

Fig. 15.1

i. State the principle of moments. [1]

ii. Calculate the minimum effort required to operate the pump. [2]

iii. State the effects of reducing the length of the effort arm. [1]

iv. State how friction can be reduced in the pump. [1]

[Total: 10]